

General Information

Department: Computer Science

Course Name: Data Structure Analysis

Course Number: CSCI 315

Credit Hours: 4 credit hours; 3 lecture hours and 2 laboratory hours.

Approved Major Credit: BS in Computer Science, BS in Cybersecurity, BA in Applied Computing, BT in Computer Science, and BT in Cybersecurity

LAC Requirement: None

Prerequisites: CSCI 325 with a grade of 'C' or better.

Course Description

The effective application of data structures and abstract data types. Abstract data types studied include lists, stacks, queues, and trees. Implementation methods include arrays, classes, pointers, and recursion. Analysis methods include Big-O notation using induction and recurrence relations. Topics also include ethical issues in computer science. (C++ is currently used.)

Motivation

This course is designed to teach you many of the basic data structures used in computer science. You will learn how to make them, use them, and analyze them. In addition, you will see how to apply them to novel problems using careful analysis of the problem to determine which data structure is best suited. Coding will be done in C++, but the concepts are not language-dependent.

This course is one of the most important in the major. Projects will require a level of design that is new to your programming career and will help to prepare you for real-world jobs. The 'toolbox' this class will provide you will be used throughout your programming future. The course serves as a gateway to many advanced courses. CSCI 315 is a *demanding* course, expect more work than previous courses. The projects in this class are more involved, requiring a deeper understanding of Computer Science. Learning to design and manage large projects is a core concept of the degree.

Course Materials

Required Textbook

Malik, D. S. (2018). *C++ Programming: Program Design Including Data Structures*. 8th edition. Stamford, CT: Cengage Learning. ISBN: 978-1-337-11756-2. *You may also choose to purchase the cheaper 7th edition for this course.*

Other Required Materials

This course relies on [GitHub](https://github.com/csu-cs/CSCI-315-2026-Fall). Assignments and lectures are located on GitHub at <https://github.com/csu-cs/CSCI-315-2026-Fall>. If you cannot access it, please email me so I can grant you access.

Student Learning Outcomes

By the conclusion of this course, students will be able to:

#	Student Learning Outcome	ABET-CAC SLO Alignment
1	Navigate a Unix/Linux environment using command-line interface tools, compile C++ programs using compiler tools (like GCC/Clang and Makefiles), and manage source code repositories using Git.	2, 6
2	Allocate, manage, and deallocate memory dynamically in C++, and utilize diagnostic tools to detect and resolve memory leaks, segmentation faults, and pointer errors.	2, 6
3	Design and execute automated unit tests that cover normal, edge, and failure cases to verify software correctness and prevent regressions.	2
4	Implement robust software components in C++ using core object-oriented principles, including classes, encapsulation, inheritance, operator overloading, and templates (parametric polymorphism).	2, 6
5	Quantify and compare the computational efficiency of algorithms and data structures analytically using Big-O notation, recurrence relations, and induction, as well as empirically through execution timing.	1, 6
6	Implement and manipulate basic linear data structures, including singly-linked lists, doubly-linked lists, stacks, and queues.	2, 6
7	Implement and manipulate hierarchical and specialized trees, including Binary Search Trees (BST), Heaps, B-Trees, and Quadrees.	2, 6
8	Represent graphs using adjacency lists and adjacency matrices, and apply graph search algorithms (Depth-First Search and Breadth-First Search) to solve pathfinding and spatial search problems.	1, 2, 6
9	Select, implement, and evaluate quadratic (bubble, selection, insertion) and divide-and-conquer (merge, quick) sorting algorithms, along with linear and binary search methods.	1, 2, 6
10	Design and implement hash tables, analyzing the performance tradeoffs between chaining and open-addressing collision resolution strategies.	1, 2, 6
11	Appropriately integrate standard containers, iterators, and generic algorithms from the C++ Standard Template Library (STL) to construct efficient software systems.	2, 6
12	Evaluate the ethical, legal, and professional implications of computing practices, particularly regarding software reliability, testing standards, and academic integrity.	4

ABET-CAC Student Outcomes Reference

The following ABET Computing Accreditation Commission (CAC) student outcomes are supported by this course. Students completing this course will have the ability to:

#	Student Learning Outcome
1	Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.
2	Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
3	Communicate effectively in a variety of professional contexts.
4	Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
6	Apply computer science theory and software development fundamentals to produce computing-based solutions.

Tentative Weekly Schedule

The weekly schedule is subject to change. **Those who keep up with the work generally do well.**

Day	Lectures	Related Assignment
1A (08/25)	Introduction and Unix/Linux Commands	Lab 01
1B (08/27)	C++ Compilation in Linux	Lab 02
2A (09/1)	Memory Management	Lab 03
2B (09/3)	Pointers and Testing	Lab 04, 05, & Ethics
3A (09/8)	C++ Object-Oriented Programming	
3B (09/10)	Operator Overloading and Templates	Lab 06
4A (09/15)	Empirical Analysis and Analytical Analysis	Lab 07
4B (09/17)	Linked Lists	Lab 08
5A (09/22)	Debugging and Doubly-Linked Lists, & the STL	Project 1 & Lab 09
5B (09/24)	Search and Quadratic Sorting	Lab 10
6A (09/29)	Quick Sort and Merge Sort	Lab 11
6B (10/1)	Stacks, Queues, Midterm Study Guide	Lab 12
7A (10/6)	Depth- and Breadth-First Search (2D Maze Search)	Project 2 & Lab 13
7B (10/8)	Midterm Exam	
8A (10/13)	Fall Break (No Class)	
8B (10/15)	Binary Search Trees: Insertion, Search and Intro to Traversal	Lab 14
9A (10/20)	Binary Search Trees: Deletion and Generic Traversal	Lab 15
9B (10/22)	Heaps & Heapsort	Lab 16
10A (10/27)	STL Containers and Iterators and STL Algorithms	Lab 17
10B (10/29)	Hash Tables via Chaining	Lab 18
11A (11/3)	Project 3 and Hash Tables via Open-Addressing	Project 3 & Lab 19
11B (11/5)	Priority Queues	Lab 20
12A (11/10)	B-Trees	Lab 21
12B (11/12)	Graphs via Adjacency Matrices	Lab 22
13A (11/17)	Graphs via Adjacency Lists	Lab 23
13B (11/19)	Quadrees	Lab 24
14A (11/24)	Brainstorm Optimizations for Project 3	

Day	Lectures	Related Assignment
14B (11/26)	<i>Thanksgiving</i> (No Class)	
15A (12/1)	Comparison of Data-Structure Performance	
15B (12/3)	Final Exam (Part 1)	
16B (12/10)	Final Exam (Part 2) at 10:30 a.m. per the official exam schedule	

Online Students

Be aware that the lack of scheduled face-to-face communication puts online students at a disadvantage. Therefore, communication is paramount. I will answer email during office hours. Please keep assignment due dates in mind, because there may be a delay in my response time. In short, get started early and stay ahead!

Video Recording of Lectures

I will video-record my in-class lectures and post the links on [the course website](#). These recordings are for online students so they can review the material. Lecture videos from the previous semester are currently posted so you can work ahead.

Grading

The assessment categories are weighted as follows.

Task	Percentage
Ethics Paper	5%
Labs	36%
Projects	30%
Midterm	14%
Final	15%

Grading Scale for Letter Grade

Final letter grades will be calculated using the following ranges.

Letter Grade	Grade Range %	Grade Points per Credit
A	90% and 100%	4.0
B+	87% and < 90%	3.5
B	80% and < 87%	3.0
C+	77% and < 80%	2.5
C	70% and < 77%	2.0
D	60% and < 70%	1.0
F	less than 60%	0
FA	Failure for Absences	0
FD	Academic Dishonesty	0

Late Work

If a lab or project is submitted within 7 days of the deadline and before the last day of class, the work will earn 80% credit. After 7 days, the work will earn 0% credit. Late work will **not** be accepted after the final day of class (before finals). Exams and papers may **not** be submitted late.

Teamwork and Assistance

There are no group projects in this course. All work is to be completed individually without any assistance from students, online tutoring, generative AI, or other services. You may seek help from your professor.

Because this class is difficult, do not allow yourself to fall behind. There is a significant snowball potential. Significantly more work is required compared to previous classes. Expect to put *many* hours into the projects (10, 20, or 30 hours depending on your skill). **Do not** wait until the last moment. **Do** come to me for help during office hours. **Do** discuss your designs and ideas with me and not just your programming bugs.

Attendance

Student participation is crucial for academic success. Students are also expected to check their BucMail daily and review Blackboard for course announcements.

In-Person

Any student who misses 25% of course meetings will be awarded a grade of FA (Failure due to Absences). In this course, **FA is awarded after missing 7 class meetings**. Three (3) instances of tardiness or leaving class early are equal to one absence. For more information, please see the [CSU Excessive Absence Policy](#). If you arrive after the roll is called, check in with the professor directly after class so that you will be recorded as tardy instead of absent.

Online

Students are expected to log in to their course(s) daily, watch lecture videos, and complete readings and assignments. Students are also expected to check their BUCmail daily.

Any student who does not **participate** in this course's academic activities for 28 consecutive days will be awarded a grade of FA (Failure due to Absences). Academic activities include completing assignments, quizzes, and exams (simply logging in does **not** count as attendance). For more information, please see [CSU's Excessive Absences policy](#).

Course, Department, and University Policies

Academic Integrity and the Honor Code

All students are expected to adhere to [Charleston Southern University's Academic Integrity Policy](#) and the [Computer Science Departmental Guidelines](#). **All assignments** are individual assignments unless explicitly specified by the professor. Do not collaborate, search for posted solutions, or post code online. **Make sure that you write every line of your own code.** You should not use **ANY** outside sources of code. Referencing code written by someone else (including AI) or sharing your code with others (online or in-person) is considered a violation of the Academic Integrity Policy and will be reported to the registrar's office. **Publicly posting code related to assignments is prohibited** (e.g., **don't** post to forums, blogs, public repositories, chegg.com, etc.). Do NOT look at your neighbor's screen for hints or ask, "how did you do that?", unless you talk to me **beforehand**.

- **NEVER** look at someone else's code in person or online (chegg.com, forums, email, etc.).
Do ask your professor if you have questions or get stuck.
- **NEVER** search online for assignment solutions.
Do reference code from the book, code given to you by the instructor, and online documentation on the C++ language.
- **NEVER** exchange code in any manner, or you tell someone what code they need.
You may talk to your classmates about C++ or assignments if you are not sharing ideas for assignment solutions.
- **NEVER** use an AI tool to generate code for use in an assignment or exam without explicit permission from the instructor for that particular task. As an example, Visual Studio Code extensions that generate code (like GitHub Copile, Tabnine, and Cody) are prohibited.
You may use AI to help explain concepts or study before exams, but not for any use for planning, debugging, developing, or any other use for assignments, exams, or other assessments.

AI Policy for CSCI 315

Level 1: Use of generative AI is *prohibited* in this course.

To ensure mastery of the concepts and skills in this course, the use of generative artificial intelligence (AI) tools is prohibited. Students who are unsure of this policy or any assignment-specific directions, including whether a given technology is considered AI, should consult the course instructor before using such technology to complete their assignment.

While proper AI use is an important skill that will be emphasized in other courses, AI misuse short-circuits the learning process to give the illusion of proficiency without the necessary depth. It is unethical to take full credit for work created with the help of AI. This principle is the same as using someone else's idea without citing it.

[Review the departmental policies for AI use in Computer Science coursework.](#)

[Click here for additional guidance on academic integrity for Computer Science coursework.](#)

A Community of Honor

As a liberal arts university committed to the Christian faith, Charleston Southern University seeks to develop ethical men and women of disciplined, creative minds and lives, focusing on leadership, service, and learning. The Honor System of Charleston Southern University is designed to provide an academic community of trust in which students can enjoy the opportunity to grow both intellectually and personally. For these purposes, the following rules and guidelines will be applied.

Academic Dishonesty

"Academic Dishonesty" is the transfer, receipt, or use of academic information, or the attempted transfer, receipt, or use of academic information in a manner not authorized by the instructor or by university rules. It includes, but is not limited to, cheating, plagiarism, and forgery as well as aiding or encouraging another to commit academic dishonesty.

"Cheating" is defined as wrongfully giving, taking, or presenting any information or material borrowed from another source (including the Internet) by a student with the intent of aiding themselves or another in academic work. This includes, but is not limited to, a test, examination, presentation, experiment, or any written assignment, which is considered in any way in the determination of the final grade. Using AI in violation of a course or assessment's stated AI protocol is cheating.

“Plagiarism” is the taking or attempted taking of an idea, a writing, a graphic, music composition, art, or datum of another person or Artificial Intelligence (AI) tool without giving proper credit and presenting or attempting to present it as one’s own, with or without intent. It is also taking written material of one’s own that have been used for a previous course assignment and using it without reference to it in its original form. Any use of AI models without proper citation is plagiarism.

Students are encouraged to ask their instructor(s) for clarification regarding their academic dishonesty standards. Instructors will include academic dishonesty/integrity standards, including explicit permissible AI use, on their course syllabi and each individual assignment.

Violations of this policy will result in academic discipline, up to and including expulsion from the University.

For more information on procedures and violation appeals, refer to the [Student Handbook](#).

Course Evaluations

To pursue our mission of *Academic Excellence in a Christian Environment*, we must receive feedback from students. The student feedback survey is online and will be available to students in the second half of the semester. Students are strongly encouraged to complete the short evaluation survey, which is anonymous. Your professor will let you know when the survey is available. The survey will be available through your MyCSU account. We greatly value your opinion!

Student Representatives

These are students who are designated by letter to represent the University in official business (e.g., athletic, music, and similar events). If officially scheduled absences cause these students to miss tests, assignments, or similar academic activities, university policy allows these to be made up without penalty. Student Representatives may opt to either make-up tests *before* departure or supplant missed tests with the final exam grade. Final exams must always be taken *before* departure to avoid an Incomplete for the course. Scheduled assignments remain subject to the lateness policy and must be turned in before departure to avoid lateness penalties. Student Representatives are responsible for informing the instructor of official absences and to make all appropriate arrangements.

Internet Etiquette

Charleston Southern University (CSU) holds students, faculty, and staff to the highest standards of conduct and expects to demonstrate courteous behaviors and practices in online communications. This policy includes guidelines and recommendations for online communications. Being respectful, thoughtful, meaningful, and ethical are fundamental to good netiquette.

CSU’s basic netiquette rules are:

- Course communications are for internal use only and are considered confidential. Do not forward or quote discussion posts, emails, or other course communications to outside parties.
- Never share personal login usernames, IDs, or passwords.
- Do not type in all capital letters. It is perceived online as shouting.
- Use proper capitalization, grammar, spelling, and punctuation conventions for professional communications.
- Avoid texting jargon or abbreviations without explanation.
 - Incorrect: “CSU is a wonderful university.”

- Correct: “Charleston Southern University (CSU) is a wonderful university.”
- Be mindful of sending emails. Ensure that content is relevant and pay attention to *Reply* versus *Reply All*.
- BucMail is the only email allowed for course communications. Other platforms (Yahoo, Gmail, etc.) are prohibited.
- In video conferencing, mute your microphone when not speaking.
- Differing views are natural and welcome in discussion boards. Be respectful in your comments, even if you disagree or dislike someone’s position on a topic.
- Respect the time and availability of students, faculty, and staff. Emails should be addressed within one business day. Remember that traditional faculty work hours are 8 a.m. - 5 p.m. EST.

Accessibility Services

Any student who may need accommodations should review the requirements/procedures on the [Accessibility Services website](#). Once approved to receive accommodations, the student must contact the instructor.

Title IX: Confidentiality and Responsible Employee Statement

Charleston Southern University is committed to maintaining a safe learning environment for everyone. In accordance with Title IX of the Education Amendments of 1972, the university prohibits any form of sexual harassment, including quid pro quo harassment, hostile environments, sexual assault, dating/domestic violence, and stalking. This policy applies to all students, employees, and visitors.

Additionally, Title IX prohibits discrimination against students based on pregnancy, childbirth, false pregnancy, termination of pregnancy, or recovery from these conditions. Pregnant or parenting students may receive accommodations to ensure their full participation in educational programs. These adjustments can be arranged through the university’s Title IX Coordinator or their designee.

Under Charleston Southern University’s Title IX Policy, all faculty members, including teaching assistants, are required to report any disclosures of sex or gender-based discrimination or violence to the Title IX Coordinator. The Title IX Coordinator will provide support and resources while maintaining privacy. If you or someone you know needs assistance, please contact the Title IX Coordinator: Shannon Morrison or the deputy Title IX Coordinator Justin Brown. Email: titleix@csuniv.edu

Please visit the [CSU Title IX webpage](#) for more information on Title IX procedures, anonymous reporting, or available support. You may complete an incident report using the form on the title-ix website.

See all course, department, and university policies located in [Blackboard](#) and the [CSU Student Handbook](#).